Gerard Chaparro Redondo

Practica 3 ESC2

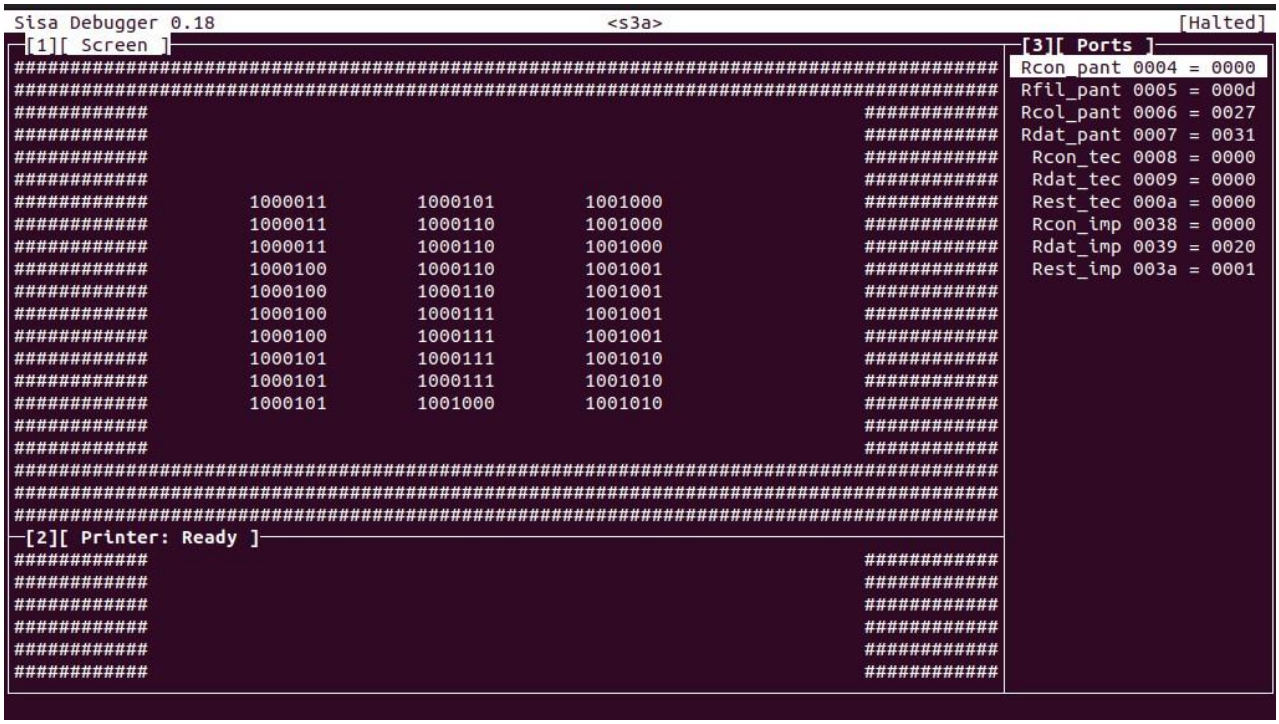
Codi:

```
.include "macros.s"
.include "crt0.s"

.data
X: .word 1,2,3,4,5,6,7,8,9,10
Y: .word -2,0,3,6,-2,12,-3,14,5,6
Z: .word 0,0,0,0,0,0,0,0,0,0
col1: .word 15
col2: .word 30
col3: .word 45
fil: .word 4
.balign 2
.text
main:
Movi R1,0
$Movei R2,X
$Movei R3,Y
$Movei R4,Z

for: LD R5,0(R2)
     LD R6,0(R3)
     AND R6,R6,R5
     ST 0(R4),R6
$Movei R6,-3
SHL R5,R2,R6
$Movei R6,0x01FF
AND R5,R5,R6
$Movei R6,-12
SHL R6,R2,R6
#mostramos el line_cache por pantalla
$Movei R0,fil
LD R0,0(R0)
out Rfil_pant,R0
while1: Movi R0,1
        AND R6,R5,R0
        $Movei R0,48
        ADD R6,R6,R0
        $Movei R0,col1
        LD R0,0(R0)
        out Rcol_pant,R0
        out Rdat_pant,R6
        $Movei R0,0x8000
        out Rcon_pant,R0
        $Movei R0,-1
        SHL R5,R5,R0
        $Movei R0,col1
        LD R0,0(R0)
        ADDI R0,R0,-1
        $Movei R6,col1
        ST 0(R6),R0
        BNZ R5,while1
$Movei R6,col1
Movi R0,15
```

```
ST 0(R6),R0
$Movei R6,-3
SHL R5,R3,R6
$Movei R6,0x01FF
AND R5,R5,R6
$Movei R6,-12
SHL R6,R3,R6
#mostramos el line_cache por pantalla
$Movei R0,fil
LD R0,0(R0)
out Rfil_pant,R0
while2: Movi R0,1
        AND R6,R5,R0
        $Movei R0,48
        ADD R6,R6,R0
        $Movei R0,col2
        LD R0,0(R0)
        out Rcol_pant,R0
        out Rdat_pant,R6
        $Movei R0,0x8000
        out Rcon_pant,R0
        $Movei R0,-1
        SHL R5,R5,R0
        $Movei R0,col2
        LD R0,0(R0)
        ADDI R0,R0,-1
        $Movei R6,col2
        ST 0(R6),R0
        BNZ R5,while2
$Movei R6,col2
Movi R0,30
ST 0(R6),R0
$Movei R6,-3
SHL R5,R4,R6
$Movei R6,0x01FF
AND R5,R5,R6
$Movei R6,-12
SHL R6,R4,R6
#mostramos el line_cache por pantalla
$Movei R0,fil
LD R0,0(R0)
out Rfil_pant,R0
while3: Movi R0,1
        AND R6,R5,R0
        $Movei R0,48
        ADD R6,R6,R0
        $Movei R0,col3
        LD R0,0(R0)
        out Rcol_pant,R0
        out Rdat_pant,R6
        $Movei R0,0x8000
        out Rcon_pant,R0
        $Movei R0,-1
```

```
        SHL R5,R5,R0
        $Movei R0,col3
        LD R0,0(R0)
        ADDI R0,R0,-1
        $Movei R6,col3
        ST 0(R6),R0
        BNZ R5,while3
$Movei R6,col3
Movi R0,45
ST 0(R6),R0
$Movei R0,fil
LD R0,0(R0)
ADDI R0,R0,1
$Movei R6,fil
ST 0(R6),R0

ADDI R2,R2,2
ADDI R3,R3,2
ADDI R4,R4,2
ADDI R1,R1,1
Movi R0,10
$CMPLT R0,R1,R0
$Movei R6,for
JNZ R0,R6
HALT
```

Prova de funcionament:



Quant els tinguis traudit comprova en el simulador les adreces assignadesals vectors X, Y i Z. Amb l'adreça inicial de X, Y, Z dedueix les adreces de:

@X[0]= 021a @X[9]= 022c

@Y[0]= 022e @Y[9]= 0240

@Z[0]= 0242 @Z[9]= 0254

Per veure on es troben:

```
Sisa Debugger 0.18                                    <s3a>                                    [Halted]
[1][ Disassembler: s3a.s ]
                        0212  5cfa  movi    r6, -6          <0xfa>
                        0214  5d00  movhi   r6, 0           <0x00>
                        0216  a181  jnz     r0, r6
                        0218* ffff  halt
                <.data> 021a  0001  and     r0, r0, r1
                        021c  0002  and     r0, r0, r2
                        021e  0003  and     r0, r0, r3
                        0220  0004  and     r0, r0, r4
                        0222  0005  and     r0, r0, r5
                        0224  0006  and     r0, r0, r6
                        0226  0007  and     r0, r0, r7
                        0228  0008  or      r0, r0, r0
                        022a  0009  or      r0, r0, r1
[2][ Data: hexadecimal byte ]
0190   02 51 00 30 3f 20 58 5c 02 5d 80 41 e8 6b 58 5c    Q 0? X\ ] A kX\
01a0   02 5d 1e 50 80 41 fd 5c ff 5d 3e 0b ff 5c 01 5d    ] P A \ ]>  \ ]
01b0   46 0b f4 5c ff 5d 3e 0d 5c 50 02 51 00 30 05 71   F  \ ]> \P Q 0 q
01c0   01 50 40 0d 30 50 00 51 a0 0d 5a 50 02 51 00 30    P@ 0P Q  ZP Q 0
01d0   06 71 07 7d 00 50 80 51 04 71 ff 50 ff 51 78 0b    q } P Q q P Qx
01e0   5a 50 02 51 00 30 3f 20 5a 5c 02 5d 80 41 e8 6b   ZP Q 0? Z\ ] A k
01f0   5a 5c 02 5d 2d 50 80 41 5c 50 02 51 00 30 01 20   Z\ ]-P A\P Q 0
0200   5c 5c 02 5d 80 41 82 24 c2 26 02 29 41 22 0a 50   \\ ] A $ & )A" P
0210   40 10 fa 5c 00 5d 81 a1 ff ff 01 00 02 00 03 00   @  \ ]
0220   04 00 05 00 06 00 07 00 08 00 09 00 0a 00 fe ff
0230   00 00 03 00 06 00 fe ff 0c 00 fd ff 0e 00 05 00
0240   06 00 00 00 00 00 03 00 04 00 04 00 04 00 05 00
0250   08 00 01 00 02 00 0f 00 1e 00 2d 00 0e 00 00 00              -
[4][ Registers ]
R0:0000 R1:000a R2:022e R3:0242 R4:0256 R5:0000 R6:00fa R7:0000 PC:0218
S0:0000 S1:0000 S2:0000 S3:0000 S4:0000 S5:003a S6:0000 S7:0000
F0: 0.00000e+00 F1: 0.00000e+00 F2: 0.00000e+00 F3: 0.00000e+00
F4: 0.00000e+00 F5: 0.00000e+00 F6: 0.00000e+00 F7: 0.00000e+00
[3][ Source: s3a.s ]
ADDI R0,R0,1
$Movei R6,fil
ST 0(R6),R0

ADDI R2,R2,2
ADDI R3,R3,2
ADDI R4,R4,2
ADDI R1,R1,1
Movi R0,10
$CMPLT R0,R1,R0
$Movei R6,for
JNZ R0,R6
HALT
```